Exponential growth and decay

1

a

i Exponential growth ii 60%

b

i Exponential decay ii 80%

c

i Exponential growth ii 105%

d

i Exponential decay ii 22%

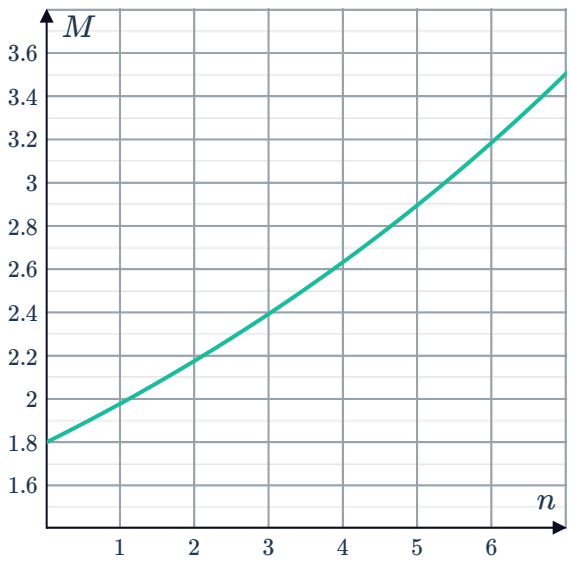
2

a $M = 2.4$

b

Months (n)	0	1	2	3	4	5	6
Mass (M)	1.8	2.0	2.2	2.4	2.6	2.9	3.2

c



3

a 440 Hz **b** 262 Hz **c** $n = 73$

4

a $y = 1300\,(1 + 0.08)^v$ **b** \$2806.60

5

a $y = 322\,000\,(0.95)^m$ **b** \$249 157.46

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6

a $y = 21\,(0.5)^n$

b 0.66 yd

7

a $A = 300\,(0.5)^{\frac{n}{20}}$

b 0.59 g

8

a $1105\ \text{km}^2$

b $332\ \text{km}^2$

9

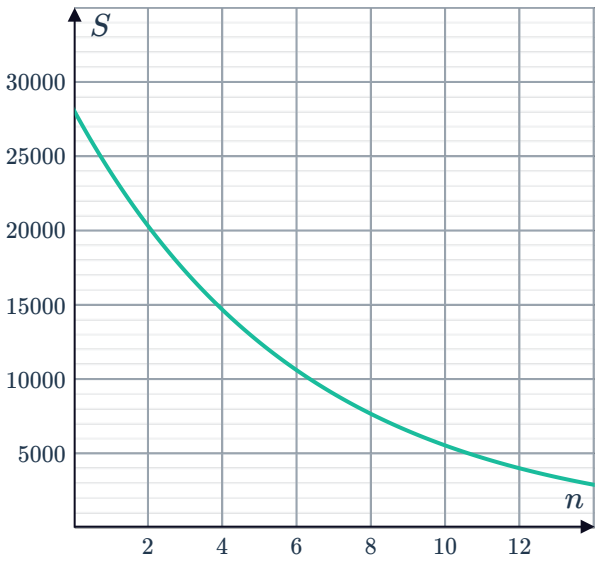
a \$900.90 **b** \$819.82 **c** $A = 990\,(91\%)^t$ **d** \$319.25

10

a

n	2	4	6	8	10	12
S	20 230	14 616.18	10 560.19	7 629.73	5 512.48	3 982.77

b



c \$17 195.5

d 9 years

11

a 0.01

b $A = 38\,200 \times 0.99^n$

c \$35 964.54

d motorbike

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12

a Decreasing

b
i No **ii** Yes **iii** Yes **iv** No

c The initial temperature of the substance

d $T = 145 \times 0.6^t$

e $0.88\,^\circ\text{C}$

f No

g Substance P

13

a $P = (0.85)^t$ **b** Maximilian **c** **C** **d** 13%

e No

Population growth

14

a 8 million

b
i 7130 million **ii** 9286 million

15 $155\,432$ people

16

a 1000 aphids **b** $32\,000$ aphids **c** 250 aphids

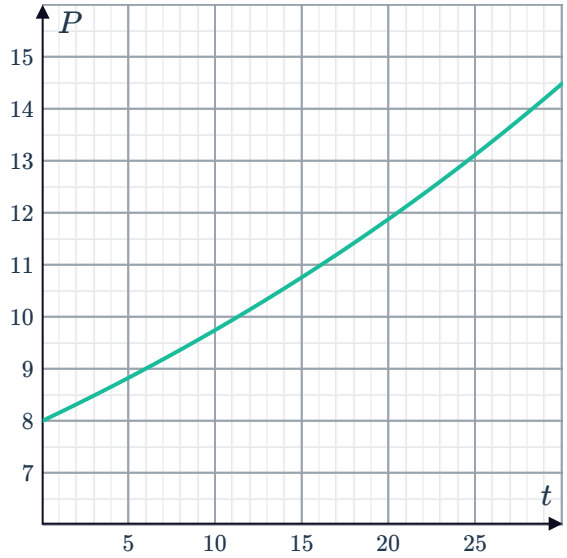
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17

a

t	5	10	15	20	25	30
P	8.8	9.8	10.8	11.9	13.1	14.5

b



c 23 years

18

a 41 199 people

b 21 669 people

19 39 650 686 people

20 233 people

21 631 bacteria

22 6591 bees

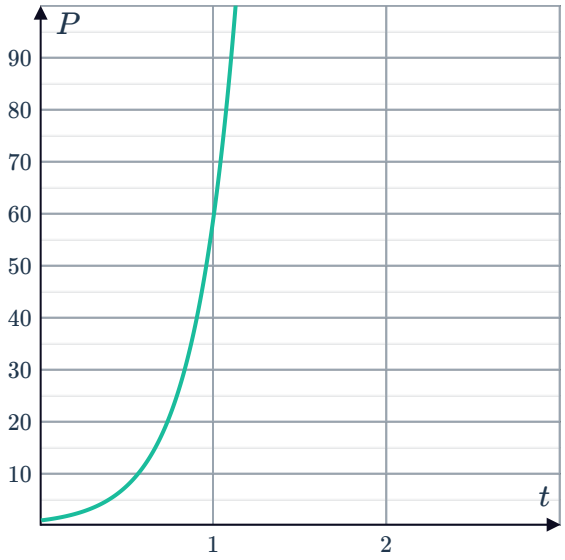
23 262 cows

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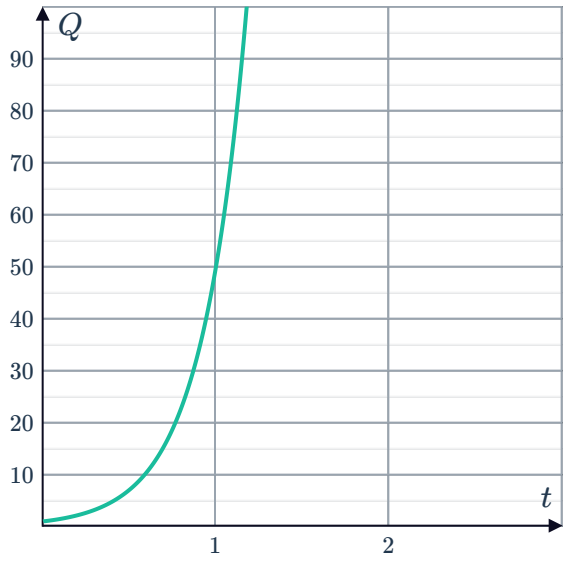
24

a Faster

b



c



d No

25

a Yes, there will be a food shortage.
We can derive the equation $F = 3\,374\,000\,(1.02)^n$ to represent the growth of the population, and the equation $P = 2\,410\,000\,(1.022)^n$ to represent the number of people the food supply will feed.
We can see that the food production is increasing, but at a slower rate than the population's rate of increase, so the population will eventually exceed the amount that can be fed.

b We can draw the graph of both functions, using technology to find any points of intersection. If the y-value of the population is greater than the y-value of the population that can be fed for the same value of n then there will be a food shortage.

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Compound interest

26 $241.70

27

a $35 730.90

b

n	1	2	3	4	5	6
A	32 190.00	35 730.90	39 661.30	44 024.04	48 866.69	54 242.02

c The value of the investment grows at an increasing rate.

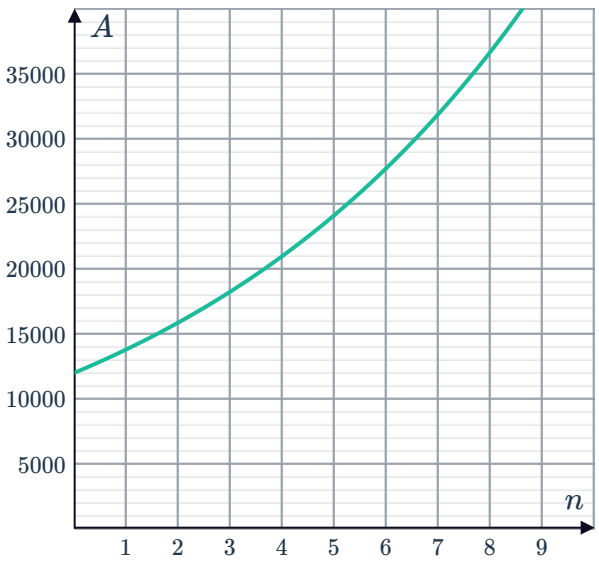
28

a $15 870.00

b

n	1	2	3	4	5	6	7
A	13 800.00	15 870.00	18 250.50	20 988.08	24 136.29	27 756.73	31 920.24

c



d $n = 4$ years

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29

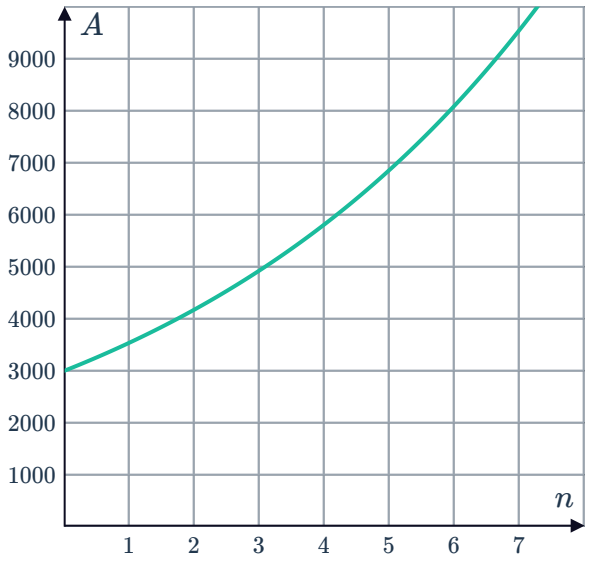
a $A = 3000 \times 1.18^n$

b \$8098.66

c

n	1	2	3	4	5	6	7
A	3540.00	4177.20	4929.10	5816.33	6863.27	8098.66	9556.42

d



30

a 0.02

b $A = 1000 \times 1.02^n$

c \$1126.16

31

a \$921

b 12

c \$35.43

32

a \$2500

b \$47 500

c \$20 000

d 10

33

a \$134 088.90

b \$24 088.90

34

a

i $A = P\,(1.053)^t$

ii $A = P\,(1.0125)^{4t}$

b 5.09%

c Bank A

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35

a $1 614 774.00

b $1 464 177.00

36

a Plan 1: $1219.92
Plan 2: $1291.17
Plan 3: $1248.25

b Plan 2

37 $0.29

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